

STAT 481 Project

MLR Analysis of Cereal Ratings

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1 Data Introduction and Description

1.1 Dataset Overview

The dataset contains nutritional information for 77 commercial breakfast cereals, each with a numeric Consumer Reports rating. Observations with missing values (coded as -1) were removed, resulting in 76 complete cases. The goal of this analysis is to predict cereal ratings using seven predictors: protein (x_1), sodium (x_2), fiber (x_3), sugars (x_4), vitamins (x_5), weight (x_6), and cups per serving (x_7). The response variable is the rating (y).

1.2 Summary Statistics

Table 1: Descriptive Statistics for All Variables

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Protein (g)	2.51	1.06	1	2	2.5	3	6
Sodium (mg)	161.78	82.32	0	133.75	180	212.5	320
Fiber (g)	2.14	2.40	0	0.75	1.75	3	14
Sugars (g)	7.03	4.38	0	3	7	11	15
Vitamins (%)	28.62	22.25	0	25	25	25	100
Weight (oz)	1.03	0.15	0.5	1	1	1	1.5
Cups	0.82	0.23	0.25	0.67	0.75	1	1.5
Rating	42.56	14.11	18.04	32.93	40.25	50.97	93.70

Most cereals have relatively low fiber and moderate to high sugar content, while sodium levels vary widely across products. The rating variable also shows substantial variation, suggesting that multiple nutritional factors may influence perceived cereal quality.

1.3 Visual Exploration

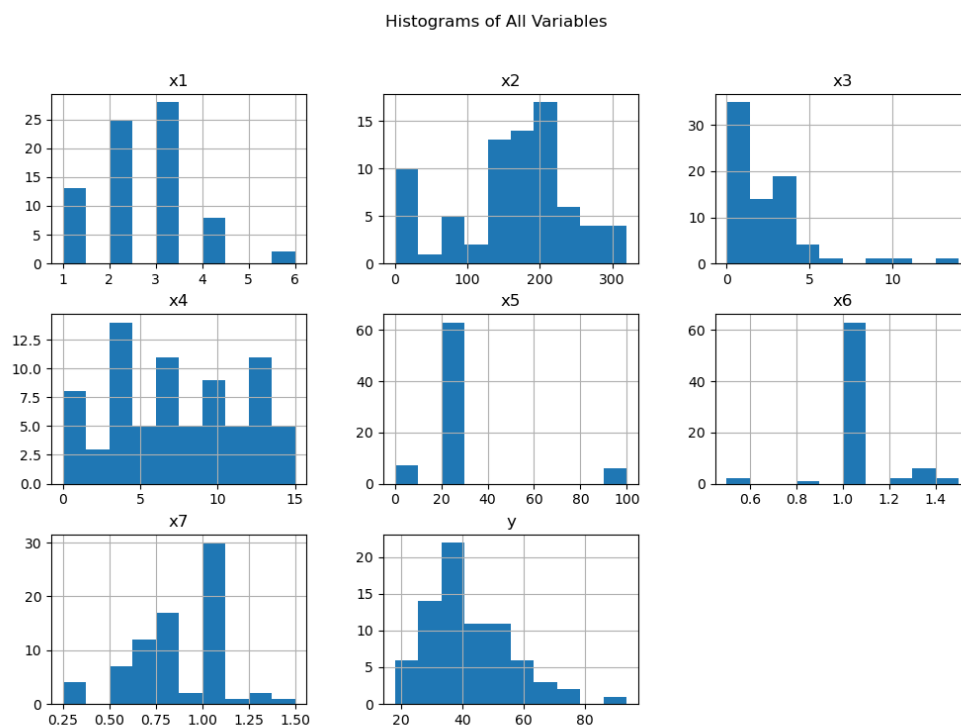


Figure 1: Histograms of all variables.

The histograms show that most cereals have relatively low fiber and sugar values, with a few cereals having much higher amounts.

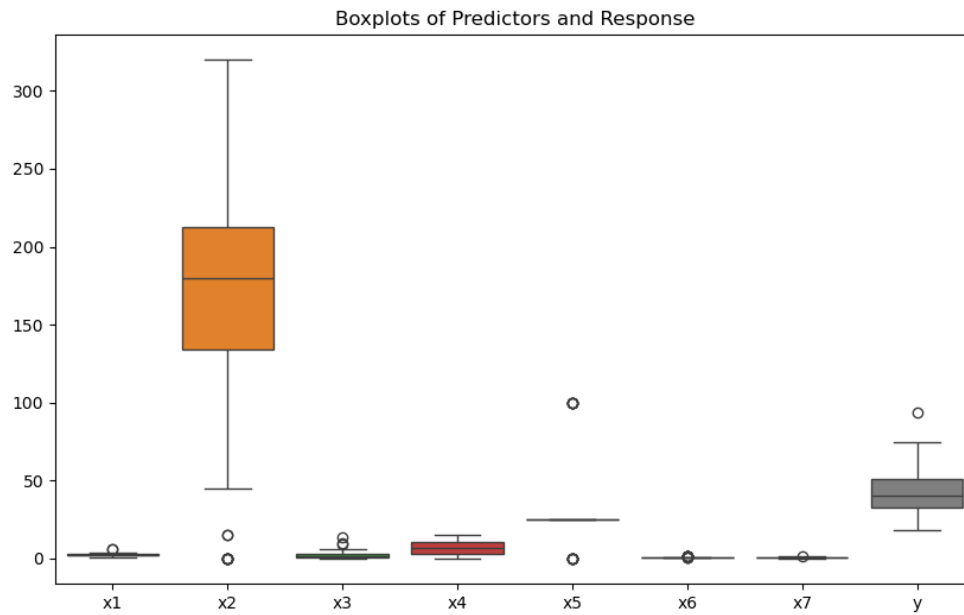


Figure 2: Boxplots of predictors and rating.

The boxplots show a few possible outliers in sodium and rating. These values were kept because they represent real cereals, not data errors.

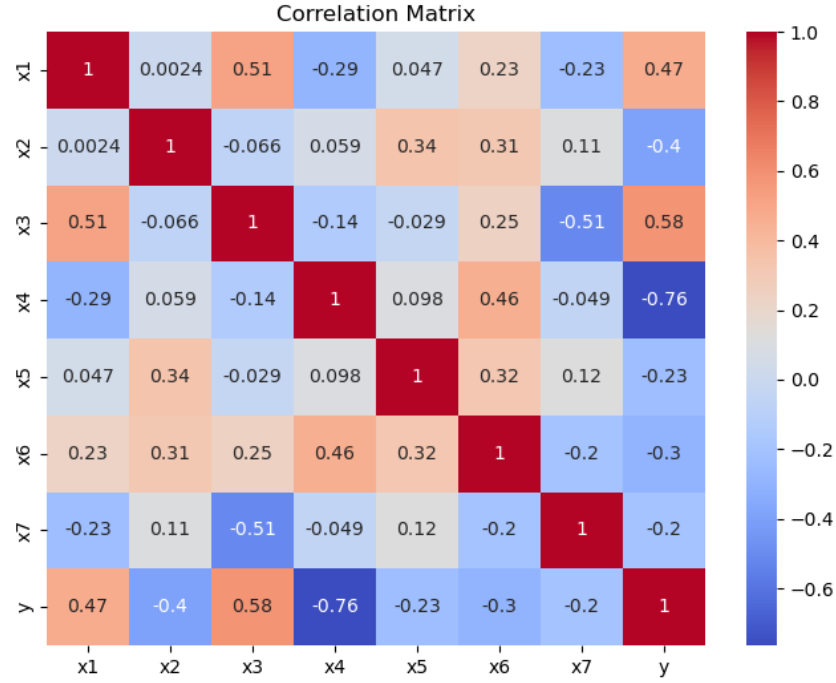


Figure 3: Correlation matrix for all variables.

Rating is strongly positively correlated with fiber and strongly negatively correlated with sugars and sodium, suggesting that these variables may play an important role in predicting cereal ratings.

2 Multiple Linear Regression Analysis

2.1 Full Model

The full multiple linear regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \varepsilon.$$

2.2 Hypotheses for Overall Model Usefulness

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_7 = 0$$

$$H_A : \text{At least one } \beta_j \neq 0$$

2.3 ANOVA Table

Table 2: ANOVA for Full Model

Source	SS	df	MS	F
Regression	7011.40	7	1001.63	127.30
Error	535.48	68	7.88	—
Total	7546.88	75	—	—

Since $F = 127.30$ with $p < 0.001$, the null hypothesis is rejected. The full model is statistically useful and explains approximately 93% of the variability in cereal ratings.

2.4 Coefficient Interpretation

Among the seven predictors, fiber has a strong positive association with rating, while sugars and sodium have strong negative associations. Protein, vitamins, weight, and cups per serving are not statistically significant after accounting for the other variables.

2.5 Multicollinearity

Variance Inflation Factor (VIF) values range between 1.2 and 2.0 for all predictors, indicating that multicollinearity is not a concern in this analysis.

2.6 Model Assumptions

Because variable selection can change the fitted model, regression assumptions are examined for both the full model and the final selected model, with primary conclusions based on the final model diagnostics.

Linearity and Equal Variance:

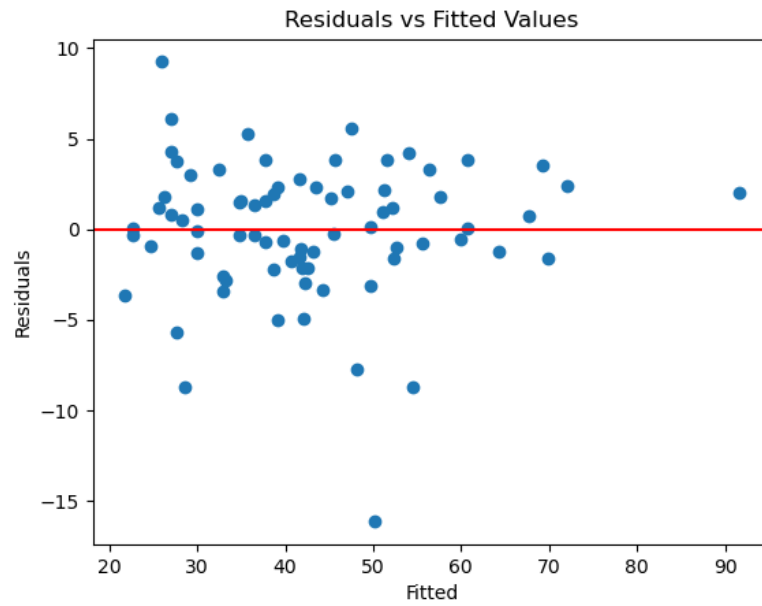


Figure 4: Residuals vs. fitted values for the full model.

The residuals show no clear systematic pattern or curvature, supporting the assumptions of linearity and constant variance. The Breusch–Pagan test further supports homoscedasticity ($p = 0.246$).

Normality:

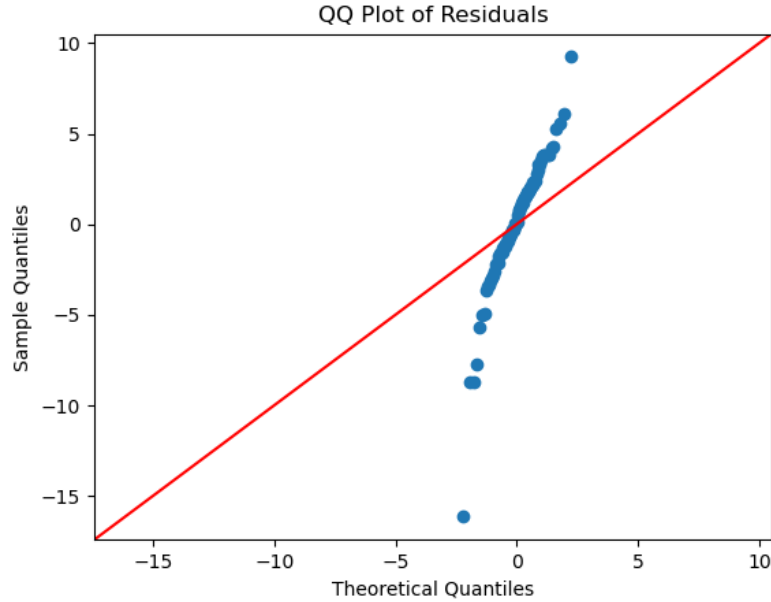


Figure 5: Q–Q plot of full-model residuals.

The Shapiro–Wilk test yields $p = 0.0003$, indicating that the residuals deviate from normality.

Independence:

The Durbin–Watson statistic is 1.80, suggesting no evidence of autocorrelation.

Addressing Non-Normality:

To address the non-normality of the residuals, a Box–Cox transformation was explored, which produced an estimated transformation parameter of $\lambda = -0.4068$. After applying the transformation, the residuals did not show a meaningful improvement in normality, and the regression coefficients became harder to interpret. In addition, the transformation did not change which predictors were statistically significant or the direction of their effects. Given the moderate sample size ($n = 76$), the lack of evidence for heteroscedasticity or autocorrelation, and the emphasis of this project on interpretability, the analysis proceeds using the original response variable, while acknowledging this limitation.

3 Model Selection

3.1 Backward Elimination

Backward elimination was performed using a significance level of $\alpha = 0.10$. Variables were removed in the following order:

$$x_6 \rightarrow x_1 \rightarrow x_5 \rightarrow x_7$$

The final model retains:

$$x_2 \text{ (sodium)}, \quad x_3 \text{ (fiber)}, \quad x_4 \text{ (sugars)}$$

3.2 Final Model

The fitted final model is:

$$\hat{y} = 61.08 - 0.0559(\text{sodium}) + 2.7525(\text{fiber}) - 2.1904(\text{sugars})$$

This model is intuitive and aligns with general nutrition expectations: cereals higher in fiber and lower in sugar and sodium tend to receive higher ratings.

3.3 Final Model Assumptions

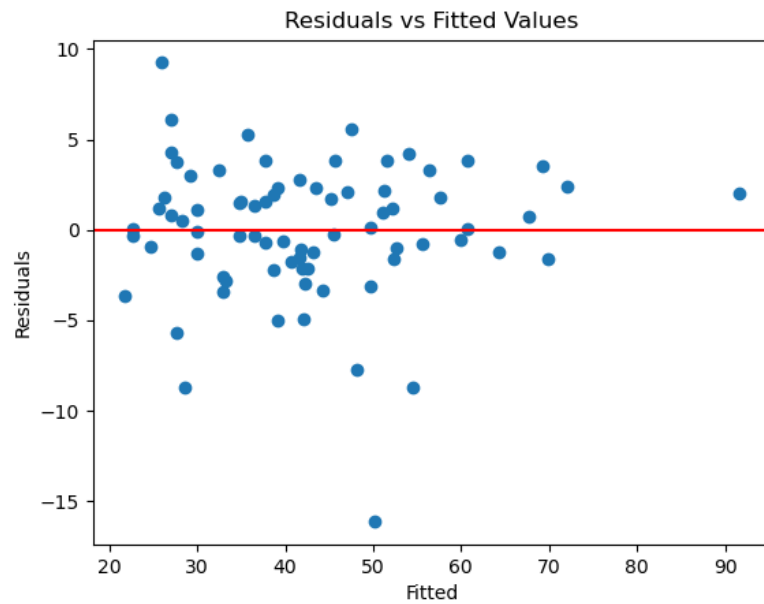


Figure 6: Residuals vs. fitted values for the final model.

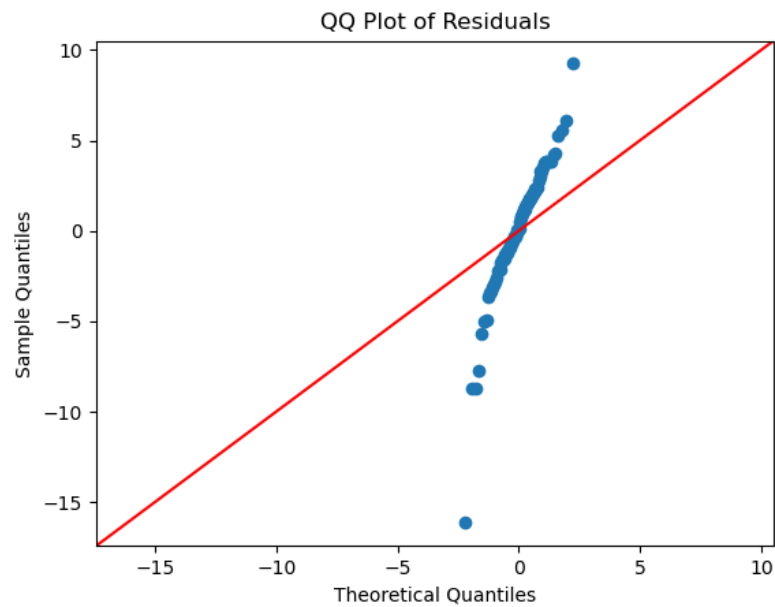


Figure 7: Q-Q plot of residuals for the final model.

Residual diagnostics for the final model show improved behavior relative to the full model. While mild deviations from normality remain, the assumptions of linearity, independence, and constant variance are satisfied. Given the skewed nature of nutritional data, the remaining non-normality is not unexpected.

4 Conclusion

This analysis shows that cereal ratings can be well explained using three key nutritional variables: sodium, fiber, and sugars. Fiber has a clear positive effect on ratings, while higher sugar and sodium content are associated with lower ratings. Compared to the full model, the final model is much simpler and easier to interpret, yet it still retains over 92% of the explained variability in ratings. While the residuals are not perfectly normal, the other regression assumptions are reasonably satisfied, and the model remains appropriate for interpretation. Overall, the results suggest that cereals higher in fiber and lower in sugar and sodium tend to receive better ratings, which aligns with general nutrition guidelines and common consumer preferences.

5 Appendix: Python Code

(Python code used for analysis is included below, as required.)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.outliers_influence import
    variance_inflation_factor
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from scipy import stats
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.stattools import durbin_watson

# Load and clean data
df = pd.read_csv(r"C:\Users\Bhavi\Downloads\Data Analytics Project (
    Stat481)\CerealsRating.csv")
cols = ["protein", "sodium", "fiber", "sugars", "vitamins", "weight"
    , "cups", "rating"]
df = df[cols]
df = df.replace(-1, np.nan).dropna()
df.columns = ["x1", "x2", "x3", "x4", "x5", "x6", "x7", "y"]

print("Dataset shape:", df.shape)
print(df.head())

# Descriptive statistics
print("\nSummary Statistics:")
print(df.describe())

# Histograms
df.hist(figsize=(12,8), bins=15)
plt.suptitle("Histograms of All Variables")
plt.tight_layout()
plt.show()

# Boxplots
plt.figure(figsize=(12,6))
sns.boxplot(data=df)

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plt.title("Boxplots of Predictors and Response")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Correlation matrix
plt.figure(figsize=(9,7))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm", center=0)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()

# Fit full model
full_model = smf.ols("y ~ x1 + x2 + x3 + x4 + x5 + x6 + x7", data=df
    ).fit()
print("\nFull Model Summary:")
print(full_model.summary())

# ANOVA table
anova_table = sm.stats.anova_lm(full_model, typ=2)
print("\nANOVA Table:")
print(anova_table)

# VIF for multicollinearity
X = df[["x1", "x2", "x3", "x4", "x5", "x6", "x7"]]
vif_df = pd.DataFrame()
vif_df["Variable"] = X.columns
vif_df["VIF"] = [variance_inflation_factor(X.values, i) for i in
    range(X.shape[1])]

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print("\nVIF Values:")
print(vif_df)

# Check assumptions
resid = full_model.resid
fitted = full_model.fittedvalues

# Residual plot
plt.figure(figsize=(10,6))
plt.scatter(fitted, resid, alpha=0.6)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted Values")
plt.show()

# QQ plot
sm.qqplot(resid, line='45')
plt.title("Q-Q Plot of Residuals")
plt.show()

# Normality test
sw_stat, sw_pval = stats.shapiro(resid)
print(f"\nShapiro-Wilk Test: W={sw_stat:.4f}, p={sw_pval:.4f}")

# Heteroscedasticity test
X_const = sm.add_constant(X)
bp_stat, bp_pval, _, _ = het_breuschpagan(resid, X_const)
print(f"Breusch-Pagan Test: stat={bp_stat:.4f}, p={bp_pval:.4f}")

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# Autocorrelation test
dw_stat = durbin_watson(resid)
print(f"Durbin-Watson: {dw_stat:.4f}")

# Box-Cox transformation
y_pos = df["y"] - df["y"].min() + 1
y_transformed, lam = stats.boxcox(y_pos)
print(f"\nBox-Cox lambda: {lam:.4f}")

# Backward elimination
def backward_elimination(data, response, alpha=0.10):
    predictors = [c for c in data.columns if c != response]
    step = 1

    while True:
        formula = f"{response} ~ " + " + ".join(predictors)
        model = smf.ols(formula, data=data).fit()
        pvals = model.pvalues[1:]
        max_p = pvals.max()

        print(f"\nStep {step}: max p-value = {max_p:.4f} for {pvals.
            idxmax()}")

        if max_p > alpha:
            remove_var = pvals.idxmax()
            print(f"    Removing {remove_var}")
            predictors.remove(remove_var)
            step += 1

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        else:

            print("  All variables significant. Stop.")
            break

    final_formula = f"{response} ~ " + " + ".join(predictors)
    final_model = smf.ols(final_formula, data=data).fit()

    return final_model, predictors

final_model, final_vars = backward_elimination(df, "y")

print("\n" + "="*60)
print("FINAL MODEL")
print("="*60)
print(f"Variables: {final_vars}")
print(final_model.summary())

# Check final model assumptions
resid_final = final_model.resid
fitted_final = final_model.fittedvalues

plt.figure(figsize=(10,6))
plt.scatter(fitted_final, resid_final, alpha=0.6)
plt.axhline(0, color='red', linestyle='--')
plt.title("Final Model: Residuals vs Fitted")
plt.show()

sm.qqplot(resid_final, line='45')
plt.title("Final Model: Q-Q Plot")
plt.show()

```



```

# Final model tests
sw_f, sw_p_f = stats.shapiro(resid_final)
print(f"\nFinal Model Shapiro-Wilk: W={sw_f:.4f}, p={sw_p_f:.4f}")

X_final = df[final_vars]
X_final_const = sm.add_constant(X_final)
bp_f, bp_p_f, _, _ = het_breuschpagan(resid_final, X_final_const)
print(f"Final Model Breusch-Pagan: stat={bp_f:.4f}, p={bp_p_f:.4f}")

dw_f = durbin_watson(resid_final)
print(f"Final Model Durbin-Watson: {dw_f:.4f}")

print("\n" + "="*60)
print("MODEL COMPARISON")
print("="*60)
print(f"Full Model: R  ={full_model.rsquared:.3f}, Adj R  ={
    full_model.rsquared_adj:.3f}")
print(f"Final Model: R  ={final_model.rsquared:.3f}, Adj R  ={
    final_model.rsquared_adj:.3f}")

```